=== UNIT 2: ELECTROCHEMISTRY, BATTERIES & CORROSION SCIENCE ===

>> 2.1 Electrochemical Cells, Electrode Potentials & Nernst Equation

Electrochemistry investigates chemical-electrical energy conversions and metallic degradation interfaces.
* Galvanic (Voltaic) Cell: Converts spontaneous chemical redox free energy into electrical work (e.g., Daniell Cell: Zn(s) | Zn2+(aq) || Cu2+(aq) | Cu(s)).
- Anode (Oxidation): Zn -> Zn2+ + 2 e- (Negative terminal).
- Cathode (Reduction): Cu2+ + 2 e- -> Cu (Positive terminal).
- Cell Potential: E_cell = E_cathode - E_anode.
* The Nernst Equation: Quantifies electrode and cell potentials under non-standard concentrations:
E = E^0 - [(R * T) / (n * F)] * ln(Q) = E^0 - [0.0591 / n] * log10([Products] / [Reactants]) at 298 K.
* Reference Electrodes:
- Standard Hydrogen Electrode (SHE): Primary standard assigned E^0 = 0.000 V: Pt | H2(g, 1 atm) | H+(aq, 1 M).
- Saturated Calomel Electrode (SCE): Secondary reference: Hg | Hg2Cl2(s) | KCl(sat). E = +0.2422 V vs. SHE at 25 deg C.
- Silver/Silver Chloride Electrode: Ag | AgCl(s) | KCl(sat). E = +0.199 V vs. SHE.

>> 2.2 Corrosion Mechanisms: Chemical (Dry) vs. Electrochemical (Wet)

Corrosion is the unintentional, spontaneous destruction of metals by chemical or electrochemical reaction with surrounding environment.
* Dry (Chemical) Corrosion: Direct reaction with dry gases (O2, Cl2, SO2) at elevated temperatures:
- Pilling-Bedworth Rule: Predicts whether an oxide layer is protective. P-B Ratio = Volume of Metal Oxide formed / Volume of Metal consumed:
If P-B Ratio < 1 (e.g., Na, K, Mg, Ca): Oxide film is porous, cracked, and non-protective; corrosion continues unabated.
If P-B Ratio >= 1 to 2 (e.g., Al, Cr, Ni, Ti): Oxide film is continuous, dense, non-porous, and self-passivating.
If P-B Ratio > 2 (e.g., Fe, Ta): Compressive stresses cause buckling and spalling of the oxide coat.
* Wet (Electrochemical) Corrosion: Occurs in aqueous media via separate micro-anode and micro-cathode sites:
- Anodic Reaction: Metal dissolves: M -> M^{n+} + n e-.
- Cathodic Reaction (depends on environment):
1. Hydrogen Evolution (in acidic media): 2 H+ + 2 e- -> H2 (g).
2. Oxygen Absorption (in neutral/alkaline media): O2 + 2 H2O + 4 e- -> 4 OH-.

>> 2.3 Types of Corrosion and Cathodic Protection

* Galvanic Corrosion: Two dissimilar metals in electrical contact in an electrolyte; the more active metal (lower reduction potential) corrodes rapidly while the more noble metal is protected.
* Differential Aeration Corrosion: Metal exposed to varying oxygen concentrations; poorly aerated areas become anodic and corrode, while well-aerated areas act as cathodes (e.g., pitting corrosion, waterline corrosion).
* Corrosion Protection:
- Sacrificial Anode Cathodic Protection: Structure connected to a more active sacrificial metal (Zn, Mg, Al).
- Impressed Current Cathodic Protection: External DC source drives electrons into protected structure via inert anodes (graphite, platinized titanium).

[SOLVED PROBLEM 1: Cell EMF and Equilibrium Constant via Nernst Equation]

Problem: Calculate the EMF of the following galvanic cell at 25 deg C (298 K):
Zn(s) | Zn2+(0.01 M) || Cu2+(0.1 M) | Cu(s).
Standard potentials: E^0(Zn2+/Zn) = -0.76 V, E^0(Cu2+/Cu) = +0.34 V. Also calculate the equilibrium constant K_c.
Solution Step 1: Overall Cell Reaction and Standard EMF E^0_cell:
Anode: Zn(s) -> Zn2+(aq) + 2 e-.
Cathode: Cu2+(aq) + 2 e- -> Cu(s).
Net: Zn(s) + Cu2+(aq) -> Zn2+(aq) + Cu(s). (n = 2 electrons transferred).
E^0_cell = E^0_cathode - E^0_anode = +0.34 - (-0.76) = +1.10 Volts.
Solution Step 2: Apply Nernst Equation:
E_cell = E^0_cell - [0.0591 / n] * log10([Zn2+] / [Cu2+])
E_cell = 1.10 - [0.0591 / 2] * log10(0.01 / 0.1) = 1.10 - 0.02955 * log10(0.10)
Since log10(0.1) = -1:
E_cell = 1.10 - 0.02955 * (-1) = 1.10 + 0.02955 = +1.1296 Volts.
Solution Step 3: Calculate Equilibrium Constant K_c:
At equilibrium, E_cell = 0: E^0_cell = [0.0591 / n] * log10(K_c)
1.10 = (0.0591 / 2) * log10(K_c) => log10(K_c) = 1.10 / 0.02955 = 37.225.
K_c = 10^{37.225} = 1.68 * 10^37.
Conclusion: Cell EMF is +1.130 V, and the massive K_c indicates virtually complete thermodynamic conversion.

[SOLVED PROBLEM 2: Pilling-Bedworth Ratio and Oxide Layer Passivation]

Problem: Calculate the Pilling-Bedworth Ratio for Aluminum oxidation forming Al2O3. (Al: atomic weight = 26.98 g/mol, density = 2.70 g/cm^3. Al2O3: molecular weight = 101.96

g/mol, density = 3.95 g/cm^3). Predict whether the oxide layer is protective.
Solution Step 1: Stoichiometric Reaction:
2 Al + 1.5 O2 -> Al2O3.
2 moles of Al metal produce 1 mole of Al2O3 oxide.
Solution Step 2: Volume of Al2O3 formed (V_oxide):
V_oxide = Molar Mass / Density = 101.96 g / 3.95 g/cm^3 = 25.81 cm^3 per mol of Al2O3.
Solution Step 3: Volume of Al consumed (V_metal):
2 moles of Al consumed = 2 * (26.98 g / 2.70 g/cm^3) = 2 * (9.99 cm^3) = 19.99 cm^3.
Solution Step 4: Compute Pilling-Bedworth Ratio (PB):
PB Ratio = V_oxide / V_metal = 25.81 cm^3 / 19.99 cm^3 = 1.29.
Conclusion: Since the PB ratio (1.29) falls in the optimal range between 1.0 and 2.0, the Al2O3 film is continuous, non-porous, and tightly adherent, conferring excellent corrosion passivation.

[SOLVED PROBLEM 3: Rust Formation Rate and Faraday's Law of Electrolysis]

Problem: An underground steel bioreactor tank exhibits an anodic corrosion leakage current of 0.50 Amperes for 365 days. Assuming iron dissolves as Fe -> Fe2+ + 2 e- (Fe: 55.85 g/mol, F = 96,485 C/mol), calculate: (1) Total mass of steel lost to corrosion in kilograms, (2) Mass of sacrificial Magnesium anode required to protect the tank for 5 years at 0.5A (Mg: 24.31 g/mol, n = 2).
Solution Step 1: Total Charge Transferred Q in 1 Year:
t = 365 days = 365 * 24 * 3600 = 31,536,000 seconds.
Q = I * t = 0.50 A * 31,536,000 s = 15,768,000 Coulombs.
Solution Step 2: Mass of Iron Dissolved via Faraday's Law:
m = (M * I * t) / (n * F) = (55.85 * 15,768,000) / (2 * 96,485) = 880,642,800 / 192,970 = 4,563.6 grams = 4.56 kg of steel.
Solution Step 3: Required Sacrificial Magnesium Mass for 5 Years:
t_5yr = 5 * 31,536,000 = 157,680,000 seconds.
m_Mg = (M_Mg * I * t) / (n * F) = (24.31 * 0.50 * 157,680,000) / (2 * 96,485) = 1,916,588 / 192,970 = 9.93 kg of pure Mg.

[SOLVED PROBLEM 4: Potentiometric pH Measurement via Glass Electrode]

Problem: A glass electrode coupled to a saturated calomel electrode (SCE, E_ref = +0.242 V) is immersed in a fermentation broth at 25 deg C. The measured cell EMF is +0.088 V. If the glass electrode calibration equation is E_cell = E'_glass - 0.0591 * pH - E_SCE (where E'_glass = +0.750 V), calculate the exact pH of the fermentation broth.
Solution Step 1: Setup Potentiometric Equation:
E_cell = E'_glass - 0.0591 * pH - E_SCE.
Solution Step 2: Substitute Known Numerical Values:
0.088 = 0.750 - 0.0591 * pH - 0.242.

0.088 = 0.508 - 0.0591 * pH.
Solution Step 3: Solve for pH:
0.0591 * pH = 0.508 - 0.088 = 0.420.
pH = 0.420 / 0.0591 = 7.107 ~ 7.11.
Conclusion: The fermentation broth operates at near-neutral physiological pH 7.11.

>> 2.4 University Review & Examination Questions

* Q1: State the Pilling-Bedworth rule and explain its significance in determining the protective nature of oxide films on metals.
Answer: The Pilling-Bedworth (PB) rule evaluates whether atmospheric oxidation forms a protective or non-protective oxide layer. PB Ratio is defined as: PB = (Volume of metal oxide formed) / (Volume of metal consumed) = [M_oxide * rho_metal] / [n * A_metal * rho_oxide]. (1) If PB < 1 (Alkali/alkaline-earth metals like Na, Mg, Ca): The volume of oxide formed is smaller than the metal consumed. The film cannot cover the surface completely, developing porous microscopic fissures that allow oxygen to continuously penetrate and consume the substrate. (2) If 1 <= PB <= 2 (Al, Cr, Ni, Ti): The oxide volume exceeds metal volume just enough to pack densely under light compressive stress. The film forms a continuous, non-porous, self-passivating diffusion barrier halting further oxidation. (3) If PB > 2 (Fe): Excessive compressive stresses cause buckling, cracking, and spalling of the oxide coat, re-exposing fresh metal.
* Q2: Explain the mechanism of differential aeration corrosion with a labeled diagram description of pitting corrosion.
Answer: Differential aeration corrosion occurs when a metallic structure is exposed to non-uniform concentrations of dissolved oxygen. The fundamental electrochemical rule: REGIONS EXPOSED TO LOWER OXYGEN BECOME ANODIC AND CORRODE; REGIONS EXPOSED TO HIGHER OXYGEN BECOME CATHODIC AND ARE PROTECTED. Mechanism in Pitting: When a particle of dirt or an imperfect coating covers a localized spot on a steel plate, oxygen diffusion to the metal underneath is restricted. The shielded area under the drop/pit becomes anodic: Fe -> Fe2+ + 2 e-. The surrounding unshielded surface with abundant oxygen acts as cathode: O2 + 2 H2O + 4 e- -> 4 OH-. The area ratio of small anode to massive cathode concentrates current density, driving rapid, intense localized digging into the plate, forming deep microscopic pits that cause sudden structural failure.

>> 2.5 Key Takeaways and Revision Points

* Nernst equation: E = E^0 - (0.0591/n) * log(Q) at 25 deg C.
* Pilling-Bedworth ratio PB = V_oxide / V_metal; protective if 1 <= PB <= 2 (e.g., Al, Cr).
* Differential aeration causes poorly oxygenated regions to act as anodes and corrode rapidly.
* Sacrificial anodes protect structures by galvanically prioritizing the degradation of more active metals (Mg, Zn).